

Annotated Source List

Adaikkan, Chinnakkaruppan, et al. “A Multimodal Approach of Microglial CSF1R Inhibition and GENUS Provides Therapeutic Effects in Alzheimer’s Disease Mice.” *BioRxiv : The Preprint Server for Biology*, Winter 2025, p. 2025.04.27.648471, pubmed.ncbi.nlm.nih.gov/40791424/, <https://doi.org/10.1101/2025.04.27.648471>.

Summary: Research Paper

The Research Paper examines the neural and cognitive effects of the CSF1R inhibitor PLX3397 in a mouse model of Alzheimer’s Disease. Although this inhibitor increases synaptic density, it also reduces neuronal locking to gamma oscillations which is linked to a change in a gene expression that is related to synapse organization. To address this neural decoupling, the researchers decided to use the GENUS treatment along with the inhibitor. The combined treatment is shown to restore gamma synchronization, reshaped gene patterns, and produced greater improvement in memory and cognitive thinking than the treatments would have done individually. Overall, the findings found that pairing microglial-targeting drugs with the non-invasive GENUS treatment improves neural function and cognitive processing much better and faster than either of those individually.

Application to Research:

The Research paper introduced a new idea of combining the GENUS treatment with other medications that cause an influx of microglial activity, which is part of the glymphatic system that is responsible for clearing out harmful proteins. This combination not only increases the efficiency of the GENUS treatment but provides significantly better results in overall memory and cognitive processes. I can use this in my research paper to expand on how the therapy will work when paired with other medications that are used for treating Alzheimer’s.

Buzsáki, György, and Xiao-Jing Wang. “Mechanisms of Gamma Oscillations.” *Annual Review of Neuroscience*, vol. 35, no. 1, 21 July 2012, pp. 203–225, <https://doi.org/10.1146/annurev-neuro-062111-150444>.

Summary: Research Paper

Gamma oscillations are crucial to the function of the brain, playing a crucial role facilitating communication between neurons and regions of the brain. Although gamma oscillations are known to be short-lived, they are detected as local field potentials, which are low-frequency electrical oscillations recorded from a small volume of brain tissue. Gamma oscillations also reflect temporal coordination of neuron firing which are crucial for brain function. More importantly however, gamma oscillations interact with slower rhythms such as theta waves in a process called as cross-frequencing coupling which aids in the synchronization between different parts of the brain, enhancing communication and playing a key role in cognitive processes such as attention, memory, and critical thinking.

Application to Research

The Research Paper provided a lot of great information regarding gamma oscillations that helped me learn more about what it is and how it really works. It also gave me a strong background about how it can help treat Alzheimer's and breaks down the crucial role it plays in gamma based light and sound based therapy. I can use this information to expand on my foundation between gamma oscillations and the importance of communication between lobes of the brain to treat Alzheimer's.

Chen, Xixi, et al. "Gamma Oscillations and Application of 40-Hz Audiovisual Stimulation to Improve Brain Function." *Brain and Behavior*, 14 Nov. 2022, <https://doi.org/10.1002/brb3.2811>.

Summary: Research Paper

Audiovisual stimulation combines both auditory and visual systems to induce gamma oscillations to improve cognitive language, motor, mood, and sleep dysfunctions as detailed by the research. The research paper breaks down gamma oscillations into three categories, slow, mid, and fast each with different Hz frequencies. All three are crucial in brain activities ranging from critical thinking to storing memory. A study revealed that short or modulated sounds that are continuously repetitive can induce a fixed frequency of neural oscillations in the brain, causing increased regional cerebral blood flow to different parts of the brain. The research identified gamma oscillations to be responsible for memory encoding and recognition, the two most critical areas that are affected in dementia related disorders such as Alzheimer's.

Application to Research:

The Research Paper had a lot of good information on how gamma oscillations can be stimulated by audiovisual stimulation from hearing or seeing repetitive frequencies that can register in the brain. This information is a great addition to my background on audiovisual treatment and how gamma oscillations can be stimulated to the brain, restoring memory encoding and recognition in Alzheimer's patients.

"Does Flashing Light Really Lower Cortical Amyloid? | ALZFORUM." *Www.alzforum.org*, www.alzforum.org/news/research-news/does-flashing-light-really-lower-cortical-amyloid.

Summary: Educational Article

Researchers in New York University's Langone Medical Center found out that 40Hz white light had a small but significant impact on the gamma rhythms in the brains of mouse models with amyloidosis. The treatment was shown to have decreased amyloid-42 production but didn't have any effect on amyloid plaques. By using evidence from Dr. Tsai's work at the MIT laboratory, they were able to figure out that an influx of microglia was able to attack the amyloid and make up for a week of daily treatment. Due to some of the mice used in the treatment disliked the flickering lights, they didn't receive as much of a benefit and some even avoided the stimulus as a whole. The researches documented that if this happens with human patients it might be hard to get any benefit from the therapy at all.

Application to Research:

The educational article included a lot of information about the current ongoing study of gamma stimulations in the brain and the countless obstacles that are in the way.

Cooperation for patients is a big part as patients with Alzheimer's especially as it has already progressed deeply show a lot of anger and reluctance towards any sort of test. With this in mind, the therapies effectiveness and overall impact might not show significant results. I could include this information in my research paper by talking about these obstacles and include some information about although the stimulation decreases the production of amyloid-42 there is visibly no change in plaque which is different from the other studies I have read.

“GENUS.” *Picower.mit.edu*, picower.mit.edu/innovations-inventions/genus.

Summary: Research Article

Dr. Tsai and Dr. Boyden from MIT started the experiment called GENUS when they discovered that gamma frequency had far less power and synchrony in Alzheimer's model mice and so they installed gamma rhythms in the brain with non-invasive light and sound which is shown to have several beneficial effects. As they continued in their study, they figured out that long-term exposure to GENUS significantly reduces neurodegeneration and preserves memory as well as improving cognitive performance. A few years later, the team was also able to discover that the 40Hz vibrations were able to reduce Alzheimer's pathology in mice models. But human participants have also shown many positive results such as reduced brain atrophy, improved sleep patterns, face-recognition and so much more that could potentially cause this treatment to be revolutionary in the field of Alzheimer's medicine.

Application to Research:

The website is from Picower Institute so the information is highly trustworthy and accurate as they were the starting force that caused the ongoing surge of research on the novel GENUS or gamma-based light and sound therapy experiment. The information also gave me a stronger background on how the therapy was discovered and the experiments that the MIT team conducted. I can use this information to expand on how the therapy was discovered and the many trials and experiments that the starting group of researchers (the MIT group led by Dr. Tsai) conducted and the results they documented.

“GENUS: Research on Sensory Stimulation to Combat Alzheimer's Disease.”

Www.youtube.com, www.youtube.com/watch?v=2W4ZdXOSXiI. Accessed 6 Nov. 2021.

Summary: Youtube Video

The video walked through the Picower Institute at MIT and Dr. Li Huei Tsai's research on the novel GENUS treatment for Alzheimer's. Dr. Tsai and her team were able to discover that the 40Hz frequency in the brain is crucial for maintaining the brain's defenses against Alzheimer's disease. In their recent studies in cell and neuron, by exposing mice to the

crucial 40Hz frequency, Dr. Tsai's team was able to enhance the gamma rhythms in the brain. This enhancement caused reductions in tau and amyloid beta protein buildups that cause blockages in the brain. Over long periods of time, the team observed a significant reduction in many biomarkers and key Alzheimer's pathology.

Application to Research:

The video was very informative and walked through not only the GENUS treatment but also Dr. Tsai's facility at the Picower Institute and her experiments that led to groundbreaking discoveries that proved how effective the treatment can be. The video was also very trustworthy and accurate as it was from the Picower Institute itself where Dr. Tsai works at. I can add this to my research paper by explaining the process that contributed to discovering this therapy and how Dr. Tsai's team was able to prove its benefits through countless experiments.

Gill, Michael. "Chronic Treatment of Alzheimer's Disease by Gamma Light and Sound Therapy." *Withpower.com*, 9 Oct. 2021, www.withpower.com/trial/phase-alzheimer-disease-11-2022-5d9cf. Accessed 21 Dec. 2025.

Summary: Study Recruitment Website:

Researchers in MIT who are trying to find human patients to work with on the novel gamma-based light and sound therapy are recruiting people who meet certain standards. They explain the purpose of the GENUS experiment, which is to combine a stimulation of light and sound in patients with mild Alzheimer's Disease and determine if the therapy can decrease harmful proteins and improve memory. Medications are also not allowed during this trial, so any prior medication for psychological diseases or Alzheimer's disease must be stopped temporarily while conducting the experiment. The researchers also explain how the novel discovery can be a revolutionary treatment for Alzheimer's as it is non-invasive and could potentially have many benefits such as a decrease in memory loss and clearance of harmful protein. Although the disease cannot reverse the progression of the disease, it can halt its progression and improve memory.

Application to Research:

The website was very helpful in learning more about the recruitment process of the disease and finding out how human sources are being tested and what requirements they have to meet. It also provides an overview of the whole process of the treatment and how patients will be treated, exposing the clinical process and how the data is being collected. I can use this in my research paper by talking about the real life applications of this process and how human patients are being tested.

Gometz, Emma. "Could Light and Sound Therapy Treat Alzheimer's?" *Science Friday*, Science Friday, 16 Aug. 2024, www.sciencefriday.com/segments/40hz-light-and-sound-therapy-alzheimers/.

Summary: Podcast

Emma Gometz goes over how brainwaves play a crucial function in our brain, specifically gamma waves which are a critical component of the brain regarding functions like thinking and memory. She explains how both human subjects and animal subjects were tested using the gamma based light and sound therapy, both yielding monumental results that could potentially shift the course of treatments for Alzheimer's. Initially they tested mice and were surprised to see that the effects were both quick and powerful as they were said to have decreased amyloid plaques within an hour of being exposed to the 40Hz gamma waves (Gometz). Other brain cells were also shown to have major changes that change the way they look, the genes they express and other factors that could be neglected during Alzheimer's. This is caused by an increase in cerebrospinal liquid as well as the increase in the function of the glymphatic system, clearing out harmful waste much faster and halting the progression of Alzheimer's.

Application to Research:

The podcast was a new and exciting way for me to learn more about my topic and it allowed me to hear their explanations and thoughts auditorily rather than just reading an article that may seem bland at times. The podcast also answered many of my questions that had been plaguing me since the beginning of my research and allowed me to develop more of a personal attachment to the topic that will be key to a great research paper moving forward.

Jessen, Nadia Aalling, et al. "The Glymphatic System: A Beginner's Guide." *Neurochemical Research*, vol. 40, no. 12, 7 May 2015, pp. 2583–2599, [pmc.ncbi.nlm.nih.gov/articles/PMC4636982/](https://pubmed.ncbi.nlm.nih.gov/articles/PMC4636982/), <https://doi.org/10.1007/s11064-015-1581-6>.

Summary: Case Study

The glymphatic system was a relatively new discovery of how macroscopic waste clearance that utilizes a system of perivascular channels, primarily formed by astrocyte cells, helps to promote the efficient clearance of proteins and metabolites found in the Central Nervous System (the brain and spinal cord). But aside from waste elimination, the glymphatic system plays a crucial role in helping distribute non-waste compounds such as glucose, lipids, amino acids, and neurotransmitters that are related to volume transmission in the brain (Nadia Aalling). The glymphatic system primarily operates during sleep, and is largely disengaged during wakefulness, reinforcing the need for sleep so that the glymphatic system can clear neurotoxic waste products including beta-amyloid plaques which lead to Alzheimer's.

Application to Research:

The Research Paper helped me understand the processes of the glymphatic system better, connecting how it correlates with protein reduction along with gamma stimulations. I can use this to make my basis on the key role that the glymphatic system plays in gamma based light and sound therapy even stronger while including more in my background about the importance of cerebrospinal fluid and other components of the glymphatic system that aid in protein reduction.

"Memories "Lost" to Alzheimer's May Be Retrievable." *Columbia University Irving Medical Center*, 21 Aug. 2017, www.cuimc.columbia.edu/news/memories-lost-alzheimers-may-be-retrievable.

Summary: Article

A study in Columbia University has discovered that memories lost in Alzheimer's may not be lost after all as they are only unreachable or restricted due to clumps of beta-amyloid plaques and tau tangles that interfere with the brain's processes. The memory you make throughout your life is never lost, but in dementia related disorders it becomes harder and harder to recall them. Due to this discovery, the use of deep brain stimulation has the possibility to recover the memories that are restricted or hard to recall in diseases such as Alzheimer's. The study that was performed in mice was able to point out the slow decaying of the ability to recall memories as many clumps and tangles in the brain interfere with its ability to communicate and function properly.

Application to Research:

The article contained a lot of key information that helped me understand how Alzheimer's affects memory and cleared up many questions that I had along the way of my research project. The information also helped me understand how gamma based light and sound therapy substitute for deep brain stimulation that helps clear out waste in the brain and "restore" pieces of memory that were previously restricted or hard to recall. I can use this information to create a section in my research paper on how memory is really affected during Alzheimer's and how gamma based light and sound therapy restores it.

Murdock, Mitchell H., et al. "Multisensory Gamma Stimulation Promotes Glymphatic Clearance of Amyloid." *Nature*, vol. 627, 28 Feb. 2024, pp. 1–8, www.nature.com/articles/s41586-024-07132-6, <https://doi.org/10.1038/s41586-024-07132-6>.

Summary: Case Study

The glymphatic system includes many components that play a crucial part in protein reduction in the brain. The clearance of these proteins allows for increased circulation to the brain, stronger connection between neurons, and reopening synapse junctions that have been blocked by protein buildup. The glymphatic system is able to reduce the exchange between the cerebrospinal fluid and the interstitial fluid which allows for it to perform a series of phagocytosis accumulation of beta-amyloid plaques and tau tangles. Studies performed by the researchers have shown that when mice were exposed to the 40Hz sensory stimulation of gamma waves, they experienced a surge in cerebrospinal fluid which in turn caused an increase in glial-mediated phagocytosis and vascular mediated clearance which cleared waste in the brain.

Application to Research:

The Research Report broke down key concepts of the glymphatic system such as its role in the clearance of brain metabolites. By exploring deep into concepts of the glymphatic system, 40Hz brainwaves, and phagocytosis involving AQP4/cerebrospinal fluid I was able to correlate the functions of the glymphatic system to the overall impacts of gamma based light and sound therapy.

News, Neuroscience. “Daily Light and Sound Therapy May Slow Alzheimer’s Decline - Neuroscience News.” *Neuroscience News*, 27 Oct. 2025, neurosciencenews.com/daily-light-sound-memory-alzheimers-29865/.

Summary: Research Study

The new noninvasive GENUS experiment that uses 40Hz to stimulate gamma waves in the brain shows long term improvement although not significant in early-onset patients. The volunteers were part of the experiment from MIT and they were tested during three time periods of their experiment, 0 months, 3 months, and 30 months. During this time period the 3 female patients were shown to have a slight reduction in tau proteins and slower cognitive decline. Increased brain-wave responsiveness to the stimulation also increased and improvement in the circadian rhythm were documented. Overall the therapy is shown to have positive impacts over courses of two years and above and it is safe to use as a daily measure with little to no side effects.

Application to Research:

This study was important in my research as it not only showed the benefits of the experiment on human patients but also exposed the limitations of it and what conditions are necessary to get any benefits from the treatment. The discussion of the direct impact that the GENUS treatment can have on Alzheimer’s pathology can also be helpful in discussing the potential impact that the disease can have on key biomarkers that are crucial to the development of Alzheimer’s disease.

Orenstein, David. “How Sensory Gamma Rhythm Stimulation Clears Amyloid in Alzheimer’s Mice.” *MIT News | Massachusetts Institute of Technology*, Mar. 2024, news.mit.edu/2024/how-sensory-gamma-rhythm-stimulation-clears-amyloid-alzheimers-0307. Accessed 1 Oct. 2025.

Summary: Research Paper

Studies in lab mice and human experiments showed that it could help reverse Alzheimer’s as well as contribute to the clearance of amyloid-beta proteins. Mitch Murdock and his team led by Li-Huei Tsai in MIT discovered the connection between gamma frequency in the brain and the glymphatic system. By experimenting on genetically modified mice, they were able to figure out that 40Hz gamma frequency getting stimulated to the brain caused a reduction in amyloid levels while also seeing an increase in cerebrospinal fluid which is a component of the glymphatic system. Aside from increasing glymphatic flow, gamma waves are also shown to increase the pulsation of neighboring blood vessels, improving circulation in the brain and retaining more memories for longer.

Application to Research:

The Research Paper was part of the MIT research that was the origin to the gamma based light and sound therapy, causing it to be a great addition to my research paper making it both credible and reliable. I also learned a lot of information about the aquaporin 4 water channel, RNA sequencing data, and peptides that are crucial to understanding how gamma based light and

sound therapy works along with its impact on the brain of patients with dementia related disorders such as Alzheimer's.

Ravn, Christopher. "40 Hz Sound Gamma Therapy (Benefits, Uses & Safety)." *OptoCeutics*, 19 Aug. 2024, optoceutics.com/40-hz-sound-gamma-therapy-safety-sleep/?srsltid=AfmBOoqDBf3PJGbBskqoJ1lFQ_avILMYn_mqxEe8toHBfaHdPkqbOXeE. Accessed 24 Sept. 2025.

Summary: Scientific Report

Christopher Ravn, the head of the Revenue Operations at OptoCeutrics goes over the benefits of the new 40Hz sound therapy that uses gamma frequencies to sort of jumpstart the brain, causing improved function over long periods of time. Although side effects such as mild headaches and slight feeling of fatigue may occur, it can be mitigated by using low volume and gradually increasing it based on the user's comfort level and experience with the device. The 40Hz sound therapy provides a multitude of benefits from increased cognitive function, neuroplasticity, relaxation, and increased cognitive clarity that helps slow down or in some cases even reverse Alzheimer's disease.

Application to Research:

The information available in this scientific report helped me develop a stronger understanding and basis on gamma stimulations in the brain. I also was able to put together a lot of information regarding gamma frequencies and how it can stimulate increased activity from the brain from improving circulation to enhancing communication across different regions of the brain.

Staff, Sleep Review. "Device Uses Gamma Sound and Light Therapy to Enhance Sleep | Sleep Review." *Sleep Review*, 12 Apr. 2024, sleepreviewmag.com/sleep-treatments/therapy-devices/sound-acoustic-therapies/bed-side-device-harnesses-gamma-sound-light-therapy-enhance-sleep/. Accessed 22 Dec. 2025.

Summary: Newsletter

Adaptive Sound Technologies has recently introduced Sound+Therapy, a device that uses 40Hz gamma sound and optional synchronized light flashes to improve sleep quality while also aiding in patients with Alzheimer's. The device generates sounds at a specific pattern to align brain activity with gamma frequencies that are associated with relaxation and cognitive function. This customizable device has many features such as adjustable light and volume levels, and optional fan noise that can trigger sleep faster. Clinical studies of patients revealed that the experiment causes significant improvements in sleep maintenance, reduced awakenings, and overall increased sleep efficiency. Overall the device is a scientifically supported, non-invasive aid that enhances sleep through targeted audio and visual gamma-frequency stimulation.

Application to Research:

The Sound+Therapy device is a great implementation that can be combined with the current GENUS therapy to help patients not only sleep better and relax faster, allowing faster healing in the body, but also include additional treatment for Alzheimer's. The already efficient GENUS treatment will be further enhanced by this device and by combining two big problems into one hurdle a combination of these two technologies can have revolutionary impact in the field of Alzheimer's medicine. I can use this information in my research paper by example on my real life implementation section and talk about future prospects for instituting this experiment into day-to-day life for Alzheimer's patients.

Tarasoff-Conway, Jenna M., et al. "Clearance Systems in the Brain—Implications for Alzheimer Disease." *Nature Reviews. Neurology*, vol. 11, no. 8, 1 Aug. 2015, pp. 457–470, <https://doi.org/10.1038/nrneurol.2015.119>.

Summary: Case Study

The build-up of protein in the brain such as beta-amyloid plaques and tau tangles are the hallstone of Alzheimer's disease, so any treatment must find a way to slow down their progression or remove them. Extracellular metabolites such as these can be removed from the brain using transport across the blood-brain barrier performed by astroglial cells or through the meningeal lymphatic vessels. Soluble beta-amyloid plaques can also be removed by a variety of ways that include key glymphatic components such as cerebrospinal fluid and interstitial fluid that absorb it through the circulatory and lymphatic system. But the majority of studies show that most of the protein clearance occurs through the blood-brain barrier which consists of astrocytes and microglial cells which play a crucial role in the clearance of waste.

Application to Research:

The Research Paper provided a great deal of information regarding how protein buildup in the brain is cleared, what functions go along with it and how it impacts dementia-related disorders such as Alzheimer's. I was able to learn more about the role of the blood-brain barrier and other glymphatic processes that contribute greatly to gamma based light and sound therapy. I can use this information in my research paper by making a section about protein buildup in the brain, what it is and how it can be mitigated through the solutions offered by gamma based light and sound therapy.

TED. "Could We Treat Alzheimer's with Light and Sound? | Li-Huei Tsai | TED." *YouTube*, 7 Mar. 2022, www.youtube.com/watch?v=AS0K0XOMNwA.

Summary: Ted-Talk

Dr. Li-Huei Tsai, the head of the Tsai laboratory in MIT discovered that gamma frequencies might play a crucial role in the development and treatment of Alzheimer's disease. By running countless experiments in lab mice they were able to figure out that the gamma frequencies in mice with Alzheimer's were disrupted and had an unsynchronized rhythm as opposed to the normal 40Hz rhythm of a person without Alzheimer's. By reaching key parts of the brain such as the prefrontal cortex and the hippocampus, it had a profound impact on decreasing protein

buildup such as amyloid-beta plaques and tau tangles. Dr. Tsai's experiments also showed improved circulation that may play a key role in flushing out wastes as said by Dr. Tsai.

Application to Research:

The Ted-Talk was one of the first sources I looked at before I began researching my topic, exposing me to the novel discovery in the MIT lab regarding a possible treatment for Alzheimer's. The Ted-Talk had a great demonstration and visual images to help me fully grasp the context Dr. Tsai was talking about, helping me fully understand the basics of gamma based light and sound therapy, how it was derived, and how it alters the brain.

Turner, Raymond. Personal Interview. December 17, 2025.

Summary: Interview

In my interview with Dr. Turner I was able to learn a lot of crucial information about the novel GENUS treatment and all of the clinical trials and its success rates. I also specifically asked Dr. Turner about future aspirations for the treatment and how the results are looking so far as well as how the treatment works and why. He mentioned that most of the treatment is still shrouded in mystery but scientists know that it works and how it alters the brain, but the why behind it is still confusing to many. He also stated that there are many different hypotheses to how to work and why the treatment shows the results it does but there is not a definite answer. But the treatment is growing at a rapid pace and although a majority of it has only been treated in mice, Dr. Turner said that there are human experiments currently being conducted with more in the future to fully determine the capabilities and success rates of the treatment on human patients.

Application to Research:

Through my conversation with Dr. Turner I was able to learn more about the treatment and how it is tested as well as clarifying a lot of misconceptions and questions that I had along my research. Dr. Turner made it clear that most of the experiments are results from mice experiments so the treatment may differ in humans and also that a lot of information about why and how the treatment works are still not definite and that many theories exist.

Valerio Manippa, et al. *An Update on the Use of Gamma (Multi)Sensory Stimulation for Alzheimer's Disease Treatment*. Vol. 14, 15 Dec. 2022, <https://doi.org/10.3389/fnagi.2022.1095081>.

Summary: Case Study:

This case study examines the GENUS treatment as a promising non-pharmalogical and non-invasive approach to treat Alzheimer's disease. Gamma oscillations, specifically at 40Hz, play a crucial role in many brain functions such as cognition, memory, and neuronal communication. Preclinical studies on mice models show that the 40Hz sensory stimulation, whether auditory, visual, or multisensory, can entrain gamma oscillations which reduces amyloid and tau buildup in the brain. This in turn activates neuroprotective glial responses which are basically microglia that are responsible for clearing waste in the brain. While there are limited human studies, the results from the few experiments conducted showed that gamma entrainment via sensory stimulation is very feasible, safe, and yields many beneficial results. The review also highlights stimulation parameters

which suggest that the benefits of the treatment may differ based on age and medical condition.

Application to Research:

The case study was very helpful in learning more about the basics and backgrounds of the GENUS treatment and how it will be administered to real life patients. The potential effects that this treatment will have on the medical field for Alzheimer's is huge in that it can cause many different benefits that other drugs and treatments cannot due only using sensory stimulation. I can use this in my research paper by expanding on the background of the gamma-based light and sound therapy/GENUS treatment and how it will be administered to the public as well as its impact on people.

VJDementia. "Gamma Frequency Stimulation as a Treatment for Alzheimer's Disease." *YouTube*, 25 Mar. 2024, www.youtube.com/watch?v=ar075Ns9t14. Accessed 2 Nov. 2025.

Summary: Youtube Video:

Dr. Chan from the MIT Picower institute where the ongoing research on gamma based light and sound therapy is being conducted talks about the therapy as a potential treatment to Alzheimer's and discusses its pathology as well as how it works. She talked about how there is a decrease in gamma oscillations in patients with Alzheimer's and a decrease in the synchronization between different parts of the brain. The hypothesis of the lab is that boosting the gamma power in the brain using optogenetics in mice will eventually result in getting a better gamma rhythm and fix all of the problems associated with a disrupted gamma rhythm. Dr. Chan and her coworkers were also able to figure out that using this therapy you could reduce amyloid related pathology and clear out some of the key protein biomarkers that are the hallmarks of Alzheimer's disease.

Application to Research:

Dr. Chan's video helped me better understand the processes that go behind the GENUS therapy and how the MIT lab was able to discover it as not only was it a conversational tone that makes it easier to understand, but she also broke down the information really well as she had first-hand experience working with the therapy. Overall the video helped me see the mechanisms more clearly and I can use this in my research paper to talk about the pathology of Alzheimer's disease and how it is altered when the irregular gamma rhythms are fixed using the novel GENUS therapy.